

Development of the eye

The development of the eye involves a series of inductive interactions between neighbouring tissues in the embryonic head. These are the neuroectoderm of the forebrain (which forms the sensory retina and accessory pigmented structures), the surface ectoderm (which forms the lens and the corneal epithelium) and the intervening neural crest mesenchyme (which contributes to the fibrous coats of the eye and to tissues of the anterior segment of the eye). A broad anterior domain of neuroectoderm, characterized by the activation of several homeobox-containing transcriptional regulators, including *PAX6*, *RX*, *SIX3* and *OTX2*, develops the potential to form optic vesicles. Subsequent interactions between mesenchyme and neuroectoderm, involving expression of the secreted protein sonic hedgehog (*SHH*) at the midline, subdivides this eye-field region into bilateral domains at the future sites of the eyes (Chow and Lang 2001). Loss of *SHH* function causes holoprosencephaly and a range of malformations that can include cyclopia, due to incomplete separation of the prosencephalon (Dubourg et al 2004).

In three-dimensional culture of murine embryonic stem cells supplemented with the correct growth factors and Matrigel® (to promote basement membrane formation), but in the absence of surface ectoderm or lens epithelium, an epithelial vesicle develops and undergoes dynamic shape change to form a two-layered optic cup (Eiraku and Sasai 2012). The tissue so formed demonstrated interkinetic nuclear migration and a fully stratified architecture similar to that of postnatal eyes, including appropriate synapses. Similar results, i.e. production of retinal architecture and retinal pigment epithelial cells, have been reported using human embryonic stem cells (Nakano et al 2012, Zhu et al 2013).

The parallel process of lens determination appears to depend on a brief period of inductive influence that spreads through the surface ectoderm from the rostral neural plate and elicits a lens-forming area of the head. Reciprocal interactions that are necessary for the complete development of both tissues take place as the optic vesicle forms and contacts the potential lens ectoderm (Saha et al 1992, Nakano et al 2012, Fuhrmann 2010). The vascular tissue of the developing eye forms by local angiogenesis or vasculogenesis of angiogenic mesenchyme (Hughes et al 2000). (Accounts of the development of the eye are given in O’Rahilly (1966, 1983).)

EMBRYONIC COMPONENTS OF THE EYE

The first morphological sign of eye development is a thickening of the diencephalic neural folds at 29 days post ovulation, when the embryo has seven to eight somites. This optic primordium (eye field) extends on both sides of the neural plate and crosses the midline at the primordium chiasmatis. A slight transverse indentation, the optic sulcus, appears in the inner surface of the optic primordium on each side of the brain. During the period when the rostral neuropore closes, at about 30 days (stage 11), the walls of the neuromere diencephalon 1 (see Fig. 17.2) begin to evaginate at the optic sulcus, projecting laterally towards the surface ectoderm; by 32 days, the optic vesicles are formed. Failure of the specification and development of the optic vesicle is associated with mutation of several transcriptional regulator genes expressed in the eye field and leads to anophthalmia (absence of the eye) (Graw 2003). Each optic vesicle is surrounded by a sheath of mesenchymal cells derived from the head mesenchyme and neural crest; its lumen is continuous with that of diencephalon 1. By 31 days, regional differentiation is apparent in each of the source tissues of the eye. The optic vesicle is visibly differentiated into its three primary parts: at the junction with the diencephalon, a thick-walled region marks the future optic stalk; laterally, the tissue that will become the sensory (neural) retina forms a flat disc of thickened epithelium in close contact with the surface ectoderm; and the thin-walled part that lies between these regions will later form the pigmented layer of the retina (retinal

pigmented epithelium). The area of surface ectoderm that is closely apposed to the distal optic vesicle thickens to form the lens placode, and the mesenchymal sheath of the vesicle begins to show signs of angiogenesis. Between 33 and 35 days post ovulation, the lens placode and optic vesicle undergo coordinated morphogenesis. The lens placode invaginates, forming a pit that pinches off from the surface ectoderm to form the lens vesicle (Fig. 40.1). The surface ectoderm reforms a continuous layer that will become the corneal epithelium. The lateral part of the optic vesicle invaginates to form a cup; the inner layer (facing the lens vesicle) will become the sensory (neural) retina, and the outer layer, influenced by signals from the surrounding extraocular mesenchyme, becomes the retinal pigmented epithelium. As a result of these folding movements, what were the apical (luminal) surfaces of the two layers of the cup now face one another across a much-reduced lumen, the intraretinal space. The pigmented layer becomes attached to the mesenchymal sheath, but the junction between the pigmented and sensory layers is less firm and is the site of pathological detachment of the retina. The two layers are continuous at the lip of the cup (Fig. 40.2). The narrow part of the optic vesicle between the base of the cup and the brain forms the optic stalk. As well as the invagination of the lateral part of the optic vesicle, the ventral surface of the vesicle and distal part of the stalk similarly invaginate, forming a wide groove, the choroid (optic) fissure, through which mesenchyme and the hyaloid artery extend. These infoldings involve differential growth and cell movement, and high levels of proliferation in the inner neuroepithelial layer.

As growth proceeds, the fissure closes and the artery is included in the distal part of the stalk. The fusion process is characterized by apoptosis at the margins of the fissure. Failure of the optic fissure to close is a rare anomaly that is accompanied by a corresponding deficiency in the choroid and iris (congenital coloboma) and is often associated with microphthalmia (small eyes). Reduced growth of the optic cup caused by mutation of the homeobox gene *CHX10*, important for specification and growth of the neural retina, is one known cause of microphthalmia (Graw 2003). Anophthalmia, microphthalmia and coloboma are also associated with mutation of the *SOX2* gene.

DIFFERENTIATION OF THE FUNCTIONAL COMPONENTS OF THE EYE

The developments just described bring the embryonic components of the eye into the spatial relationships necessary for the passage, focusing and sensing of light. The next phase of development involves further patterning and phenotypic differentiation in order to develop the specialized structures of the adult organ.

The optic cup becomes patterned, from the base to the rim, into regions with distinct functions. Several secreted factors, including bone morphogenetic proteins (BMP), retinoic acid and *SHH*, and transcriptional regulators, including *PAX6* and *PAX2*, are important for specifying each region (Chow and Lang 2001, Sinn and Wittbrodt 2013). The outer layer of the optic cup remains as a thin layer of cells, which begin to acquire pigmented melanosomes and form the pigmented epithelium of the retina at around 36 days. In a parallel process, which begins before invagination, the cells of the inner layer of the cup proliferate to form a thick pseudostratified neuroepithelium, the future neural retina, over the base and sides of the cup. The peripheral region around the lip of the cup extends, and is further differentiated into the components of the prospective iris at the rim, and the ciliary body a little further back, adjacent to the neural retina (see Fig. 40.2). The development of this pattern is reflected in regional differences in the expression of various genes that encode transcriptional regulators and which are therefore likely to play key roles in controlling and coordinating

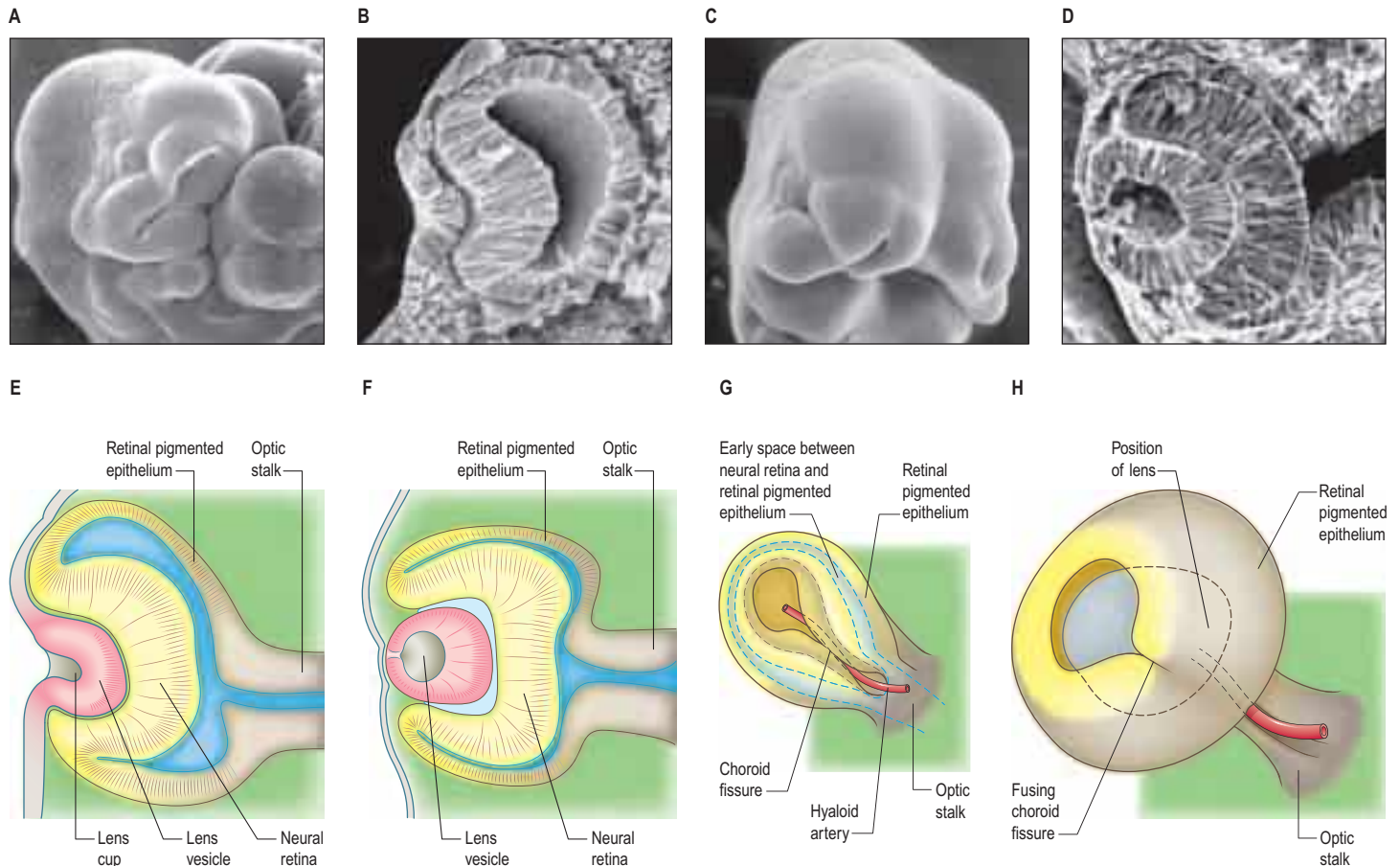


Fig. 40.1 Development and morphogenesis of the optic cup. **A–D** Scanning electron micrographs of early eye development, **A** and **C** showing external view, **B** and **D** showing longitudinal sections through the eye. **B** and **E** show the early invagination of the lens placode and the modification of the optic vesicle to an optic cup. **F** and **D** show the formation of the lens vesicle and the two layers of the retina (nervous and pigmented). **G** and **H** show three dimensional images of the outer aspect of the optic cup. **G**. Early formation of the choroid fissure along the side of the optic cup and optic stalk; the layers of the retina are shown. **H** The fused choroid fissure; the position of the lens within the optic cup is shown in dotted outline. (**A, B, C, D** Courtesy of Kathleen Sulik PhD, Professor, University of North Carolina.)

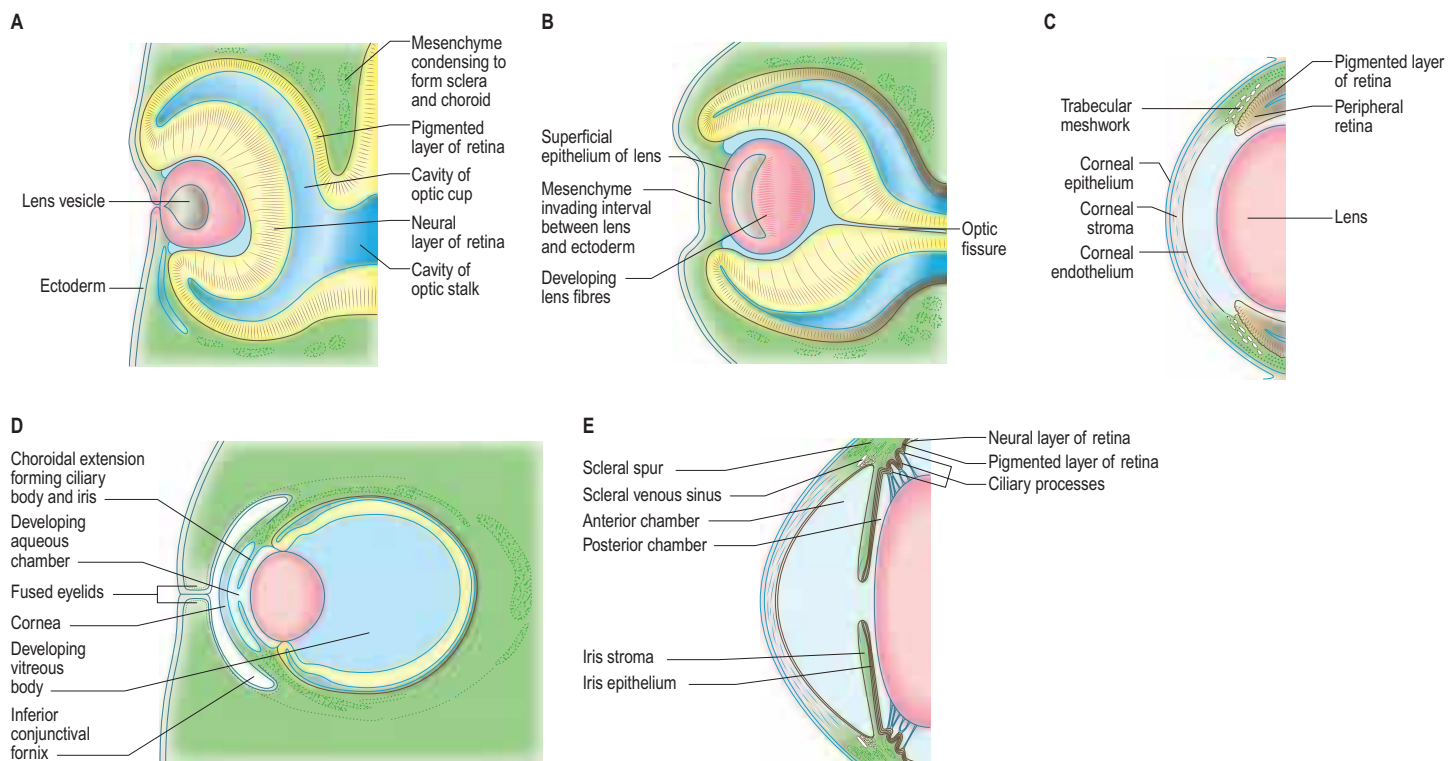


Fig. 40.2 Sections through the developing eyes of human embryos. **A**, Equivalent to 8 mm crown-rump length (CRL); stage 16. The thick nervous and the thinner pigmented layers of the developing retina and the lens are shown. The two layers of the embryonic optic cup are separated by the intraretinal space. **B**, Equivalent to 13.2 mm CRL; stage 17. The surface ectoderm anterior to the lens forms the corneal epithelium, whereas the corneal stroma and endothelium will differentiate from the invading mesenchyme (of neural crest and mesodermal origin). **C**, Equivalent to 40 mm CRL. The development of the anterior aqueous chamber is apparent with choroidal extensions and iris visible. The eyelids have developed and are fused; the extent of the conjunctival fornices can be seen. **D**, Anterior growth of the peripheral retina, pigmented layer of the retina and mesenchymal proliferation at the anterior part of the retina will give rise to the ciliary body and iris. The surface ectoderm anterior to the lens forms the corneal epithelium, whereas the corneal stroma and endothelium will differentiate from invading mesenchyme (of neural crest and mesodermal origin). **E**, Details of the developing uveal tract. Note the development of the anterior and posterior aqueous chambers, separated by the iris, and the attachment of the lens to the ciliary body.

development. Distinct sets of genes are expressed prior to and during overt cell-type differentiation. For example, *PAX6* is expressed in the prospective ciliary and iris regions of the optic cup; individuals heterozygous for mutations in *PAX6* lack an iris (aniridia), which suggests a causal role for this gene in the development of the iris. The genes expressed in the eye are also often active at a variety of other specific sites in the embryo, which may, in part, account for the co-involvement of the eye and other organs in syndromes that result from single genetic lesions, e.g. *PAX2* mutation causes coloboma and kidney defects, reflecting the sites of expression of the gene (Graw 2003).

Developing neural retina

The developing neural retina consists of an outer nuclear zone, which contains dividing neuroepithelial retinal progenitor cells, and an inner marginal zone, which is initially devoid of nuclei. At around 37 days, the cells of the nuclear zone invade the marginal zone, and by stage 18 (44 days), the nervous stratum of the retina consists of inner and outer neuroblastic layers. Cell lineage analyses have shown that seven retinal cell types are all derived from a common multipotential retinal progenitor cell. Different types of retinal cells are born (cease dividing) in a conserved sequence during development: ganglion cells, amacrine cells, cone photoreceptors and horizontal cells develop early, whereas bipolar cells, rod photoreceptors and Müller glial cells develop later (Cepko et al 1996). Newly born cells migrate from the apical (ventricular) surface to the appropriate cell layer in the developing retina, establishing its characteristic laminar structure. The developing ganglion cell layer first separates from the neuroblastic layers by formation of the inner plexiform layer. The inner nuclear layer, containing developing amacrine, horizontal, bipolar and Müller glial cells, then separates from the outer nuclear layer, containing the developing rod and cone photoreceptors by formation of the outer plexiform layer. Mature retinal neurones first appear in the central part of the retina. By the eighth month, all the named layers of the retina can be identified. However, the photoreceptor cells continue to differentiate after birth, generating an array of increasing resolution and sensitivity; the macula does not reach maturity until 15–45 months after birth (Hendrickson and Yuodelis 1984).

The divergent differentiation of the pigmented and sensory layers of the retina from the initially bipotential neuroepithelium of the optic vesicle involves activation of region-specific regulatory genes, e.g. *CHX10* in the presumptive neural retina and *MITF* in the presumptive pigmented epithelium (Bharti et al 2006). Patterning by gene expression is an important aspect of establishing regional identity of the optic cup and the subsequent maturation of these respective tissues. Soluble factors from the retina elicit the polarized distribution of plasma membrane proteins and the formation of tight junctions in the pigmented epithelium. Neural retinal differentiation is mediated by several growth factors, including fibroblast growth factors, SHH and retinoic acid. Basic helix-loop-helix proneural transcriptional regulatory genes also play a central role in regulating retinal cell fate. However, the pigmented epithelium initially retains the potential to become neural retina and will do so if the embryonic retina is wounded, demonstrating the plasticity of the early commitment to pigment epithelium or neural retinal fate.

The retinal vasculature forms by the aggregation of spindle-shaped cells (mesenchymal cells) that emanate from the optic disc by week 15 and form vascular cords, consistent with vessel formation by vasculogenesis, which give rise to the inner plexus of the retina. Vessel formation in the temporal and peripheral retina occurs by angiogenesis. New vessel segments sprout from pre-existing vessels and grow tangentially by angiogenesis into the neuroepithelium (Hughes et al 2000).

Optic nerve

The optic nerve develops from the optic stalk. The centre of the optic cup, where the optic fissure is deepest, will later form the optic disc, where the neural retina is continuous with the corresponding invaginated cell layer of the optic stalk; the developing axons of the ganglion cells, therefore, pass directly into the wall of the stalk and convert it into the optic nerve. Myelination of the axons within the optic nerve begins shortly before birth but the process is not completed until some time later. The optic chiasma is formed by the meeting and partial decussation of the axons within the two optic nerves in the ventral part of the lamina terminalis (at the junction of the telencephalon with the diencephalon in the floor of the third ventricle). Beyond the chiasma, the axons continue as the optic tracts, and pass principally to the lateral geniculate bodies and to the superior tectum of the midbrain.

Ciliary body

The ciliary body is a compound structure. Its epithelial components are derived from the region of the inner layer of the retina, between the iris and the neural retina, and the adjacent outer layer of pigmented epithelium. The cells here differentiate in close association with the surrounding mesenchyme to form highly vascularized folds that secrete aqueous fluid into the globe of the eye.

The inner surface of the ciliary body forms the site of attachment of the lens. The outer layer is associated with smooth muscle derived from mesenchymal cells in the choroid that lie between the anterior scleral condensation and the pigmented ciliary epithelium (p. 691).

Iris

The iris develops from the tip of the optic cup, where the two neuroepithelial layers remain thin and are associated with vascularized, muscular connective tissue. The muscles of the sphincter and dilator pupillae are unusual in that they are of neuroectodermal origin, and develop as a result of further growth and differentiation of the two layers of the optic cup. Mesenchymal cells, largely composed of neural crest cells that have migrated anterior to the lens, form the collagen-rich iris stroma; they overlay the pigmented epithelium of the iris, which is continuous with the ciliary body and neural retina, and is of neuroectodermal origin. The mature colour of the iris develops after birth and is dependent on the relative contributions made by the pigmented epithelium on the posterior surface of the iris and the neural crest-derived melanocytes in the mesenchymal stroma of the iris. If only epithelial pigment is present, the eye appears blue, whereas if there is an additional contribution from the melanocytes, the eye appears brown. Aniridia, the absence of the iris, is commonly caused by heterozygous mutations of *PAX6*.

Lens

The lens develops from the lens vesicle (see Figs 40.1F, 40.2A). Initially, this is a ball of actively proliferating epithelium, but by stage 16, there is a discernible difference between the thin anterior (i.e. outward-facing) epithelium and the thickened posterior epithelium. Cells of the posterior wall lengthen and fill the vesicle (see Fig. 40.2B,C), reducing the original cavity to a slit by about 44 days. The posterior cells become filled with a very high concentration of proteins (crystallins), which render them transparent; they also become densely packed within the lens as primary lens fibres. Cells at the equatorial region of the lens elongate and contribute secondary lens fibres to the body of the lens in a process that continues into adult life, sustained by continued proliferation of cells in the anterior epithelium (p. 697). The polarity and growth of the lens appear to depend on the differential distribution of soluble factors that promote either cell division or lens fibre differentiation, and are present in the anterior chamber and vitreous humour, respectively. Congenital cataracts can be associated with mutations in genes that encode structural lens proteins, particularly crystallin proteins, as well as genes that encode transcriptional regulatory factors expressed specifically in the lens, such as *MAF* and *PITX3*, which are needed for normal lens development and which also influence normal growth of the globe of the eye (Graw 2003, Graw 2010).

The developing lens is surrounded by a vascular mesenchymal condensation, the vascular capsule, the anterior part of which is named the pupillary membrane. The posterior part of the capsule is supplied by branches from the hyaloid artery, and the anterior part is supplied by branches from the anterior ciliary arteries. During the fourth month, the hyaloid artery gives off retinal branches. By the sixth month, all of the vessels have atrophied, except the hyaloid artery, which becomes occluded during the eighth month of intrauterine life, although its proximal part persists in the adult as the central artery of the retina. Atrophy of the hyaloid vasculature and of the pupillary membrane appears to be an active process of programmed tissue remodelling that is macrophage-dependent; visual impairment occurs in persistent hyperplastic primary vitreous, a condition in which normal regression of the hyaloid vasculature fails to occur. The hyaloid canal, which carries the vessels through the vitreous, persists after the vessels have become occluded. In the neonate, it extends more or less horizontally from the optic disc to the posterior aspect of the lens, but when the adult eye is examined with a slit-lamp, it can be seen to follow an undulating course, sagging downwards as it passes forwards to the lens. With the loss of its blood vessels, the vascular capsule disappears and the lens becomes dependent for its nutrition on diffusion via the aqueous and

vitreous humours. The lens remains enclosed in the lens capsule, a thickened basal lamina derived from the lens epithelium. Sometimes, the pupillary membrane persists at birth, giving rise to congenital atresia of the pupil.

Vitreous body

The vitreous body develops between the lens and the optic cup as a transparent, avascular gel of extracellular substance. The precise derivation of the vitreous remains controversial. The lens rudiment and the optic vesicle are, at first, in contact, but they draw apart after closure of the lens vesicle and formation of the optic cup, and remain connected by a network of delicate cytoplasmic processes. This network, derived partly from cells of the lens and partly from those of the retina, is the primitive vitreous body. At first, these cytoplasmic processes are connected to the whole of the neuroretinal area of the cup but, later, they become limited to the ciliary region, where, by a process of condensation, they form the basis of the suspensory ligaments of the ciliary zonule. The vascular mesenchyme, which enters the cup through the choroidal fissure and around the equator of the lens, associates locally with this reticular tissue and thus contributes to the formation of the vitreous body.

Anterior segment

Mesenchymal cells of neural crest origin migrate anteriorly around the optic cup and between the surface ectoderm and the lens to contribute to the development of anterior segment structures, including the ciliary body, the iris, the cornea and the iridocorneal angle tissues (see Fig. 40.2C–E) (Gould et al 2004). The anterior chamber initially appears as a cleft in this mesenchymal tissue. The mesenchyme superficial to the cleft forms the stroma (substantia propria) and endothelium of the cornea, and that deep to the cleft forms the stroma of the iris and the pupillary membrane. Tangentially, this early cleft extends as far as the iridocorneal angle, where communications are established with the scleral venous sinus (canal of Schlemm). Mesenchymal cells of neural crest origin lying at the angle of the anterior chamber differentiate to form a specialized meshwork of trabecular beams (collagen fibrils covered by cells); the open spaces of the meshwork become open to the anterior chamber as the beams develop. The canal of Schlemm develops deep to the trabecular meshwork and is derived from mesodermal mesenchyme. Initially a vascular structure lined by endothelial cells, the canal acts as an aqueous sinus from the fifth month of gestation (McMenamin 1989). The forward-growing optic cup rim differentiates into the ciliary epithelium and the iris; the posterior chamber is formed between the iris, the lens capsule, the zonular suspensory fibres and the ciliary processes. The ciliary processes produce the aqueous humour that flows through the pupil and is drained in the iridocorneal angle, mainly by the trabecular meshwork and the canal of Schlemm. In this way, the walls of the anterior and posterior segment chambers furnish both the sites of production, and the channels for circulation and reabsorption, of the aqueous humour (p. 687). The *FOXC1* and *PITX2* transcriptional regulatory genes are expressed in the neural crest cells migrating into the presumptive anterior segment, and regulate differentiation of the anterior segment tissues. Anterior segment dysgenesis, involving malformation of the iris, cornea and angle, occurs when these genes are mutated (e.g. Axenfeld–Rieger syndrome) and is often associated with raised intraocular pressure and glaucoma (Gould et al 2004).

Cornea

The cornea is induced in front of the anterior chamber by the lens and optic cup. The corneal epithelium is formed from surface ectoderm. The primordial corneal endothelium lining the front of the anterior chamber is formed from mesenchymal cells derived from the neural crest (see Fig. 40.2C,D,E). Mesenchymal cells migrate between these layers and differentiate to form specialized fibroblasts (keratocytes) that secrete the extracellular matrix of the corneal stroma. A regular array of collagen fibres (lamellae) is established between these two layers and serves to reduce scattering of light entering the eye. The most anterior region of the stroma (Bowman's layer) develops as an acellular zone packed with collagen fibrils that confer strength on the layer. From the third month, the endothelium is organized as a monolayer of cells and develops a strong, laminated basal lamina (Descemet's membrane) adjacent to the stroma. The endothelium maintains corneal transparency by regulating the water content of the stroma (p. 689).

Choroid and sclera

The choroid and sclera differentiate as inner vascular and outer fibrous layers, respectively, from the neural crest mesenchyme that surrounds the optic cup; the choroid is continuous with the leptomeningeal internal sheath of the optic nerve, and the sclera is continuous with the outer dural sheath of the optic nerve. The blood vessels of the choroid develop from the fifteenth week and include the vasculature of the ciliary body.

DIFFERENTIATION OF STRUCTURES AROUND THE EYE

Extraocular muscles

Development of the extrinsic ocular muscles is complex, involving coordinated juxtaposition of specific muscle precursors, cranial motor nerve outgrowth and neural crest mesenchyme. During gastrulation, when the earliest cells are migrating through the primitive node to form the prechordal plate and notochordal process, they transiently express myogenic markers. In stages 9 and 10 a population of prechordal mesenchyme cells migrate laterally from the lateral edge of the prechordal plate toward the unsegmental paraxial mesenchyme each side of the notochord. After neurulation is complete, bilateral premandibular, intermediate and caudal cavities develop adjacent to the neural tube. The walls of these 'head cavities' are lined by flat or cylindrical cells that do not exhibit the characteristics of a germinal epithelium; they were previously termed preotic somites.

As the oculomotor nerve grows towards the developing eye, at the level of the premandibular head cavity, the prechordal mesenchyme becomes apparent as a condensation of premuscle cells at its ventrolateral side. This later subdivides into the blastema of superior, inferior, medial and lateral recti and inferior oblique. Similar events occur in the intermediate head cavity which is associated with the trochlear nerve and premuscle cells forming superior oblique, and the caudal head cavity (abducens nerve and lateral rectus) (see Figs 12.4 and 35.7).

The early myogenic properties of the prechordal mesenchyme have been demonstrated experimentally; if transplanted into limb buds, the cells are able to develop into muscle tissue (Wachtler and Jacob 1986), however, the timing of expression of myosin heavy chain isoforms is different from that of limb myoblasts. Myotube formation starts later and progresses at a slower pace than in the limb, and coexpression of MyHC1/1st and MyHC1/2nd are both seen from the earliest stages of development in contrast to only MyHC1/1st in the limb. These early myoblasts retain their distinct differences from other skeletal muscles as they mature: they are smaller and loosely arranged, and belong to very small motor units (Porter and Baker 1996, Pedrosa-Domellöf et al 2000). Their final attachment to the eye is specified by neural crest formation of the sclera, extraocular muscle tendons and the orbit. In cases of anophthalmia the relative location and structure of extraocular muscles can be identified (Pedrosa-Domellöf et al 2000, Bohnsack et al 2011).

Eyelids

The eyelids are formed as small cutaneous folds of surface ectoderm with a core of neural crest mesenchyme (see Fig. 40.2D). During the middle of the third month, their edges come together and unite over the cornea to enclose the conjunctival sac; they usually remain united until about the end of the sixth month. When the eyelids open, the conjunctiva lining their inner surfaces and covering the scleral region of the eye fuses with the corneal epithelium. Transforming growth factor alpha (TGF- α) and several other growth factors regulate the mesenchyme–epithelium interactions and cell migration that are required for eyelid formation; keratinization is thought to play an important role in lid separation. The eyelashes and lid glands, i.e. sebaceous glands associated with the eyelashes and tarsal (Meibomian) glands, develop from ectoderm, as do the lacrimal and accessory lacrimal glands. Orbicularis oculi, which closes the eyelids, develops on each side from skeletal myoblasts from the second pharyngeal arch that invade the eyelids. The muscles that widen the palpebral fissure develop within the orbit from mesenchymal cells; the superior and inferior tarsal muscles are smooth muscle, whereas levator palpebrae superioris is striated muscle that is attached to each upper eyelid by a tendon derived from the neural crest (Plock et al 2005). (For a detailed timetable for upper eyelid development in staged human embryos and fetuses, see Byun et al (2011).)

Lacrimal apparatus

The ectodermal epithelium of the superior conjunctival fornix proliferates and gives rise to a series of tubular buds that form the alveoli and ducts of the lacrimal gland. The buds are arranged in two groups: one forms the gland proper and the other forms its palpebral process (de la Cuadra-Blanco et al 2003). The lacrimal sac and nasolacrimal duct are derived from ectoderm in the nasomaxillary groove (between the lateral nasal process and the maxillary process of the developing face) (see Fig. 36.11). The ectoderm thickens to form a solid cord of cells, the nasolacrimal ridge, which subsequently sinks into the mesenchyme and becomes canalized during the third month to form the nasolacrimal duct. The lacrimal canaliculi arise from the cranial extremity of the cord as buds that establish openings (puncta lacrimalia) on the margins of the lids. The inferior canaliculus isolates a small part of the lower eyelid to form the lacrimal caruncle and plica semilunaris.

Neonatal and infant eye

Low-birth-weight and preterm infants are at risk of developing retinopathy of prematurity, a proliferative retinopathy (Kashani et al 2014,

Hartnett et al 2014). At full term, the eye is 65% of its adult size. It grows rapidly for the first year, slightly slower until 3 years and then more slowly until puberty. There is proportionately less growth of anterior, compared to posterior, structures and the globe of the eye becomes more spherical. The neonatal lens is more spherical than that of the adult, which helps to compensate for the relative shortness of the eye. The visual acuity of neonates is estimated to be 20/400 and can reach 20/30–20/20 by 2–3 years of age; there is a general trend for infants to be far-sighted (Olitsky et al 2011). The visual pathways, lateral geniculate body and occipital visual cortex are patterned postnatally by exposure to visual stimuli. Visual loss with no structural anomaly of the eye, amblyopia, is caused by abnormal visual stimulation during infancy and early childhood (up to 6–7 years of age) (Ruiz de Zárate and Tejedor 2007). Neonatal extraocular muscle coordination is usually achieved by 3–6 months of age and persistent deviation of an eye requires evaluation. Preterm infants have reduced reflex and tear secretion, and tears may not be present with crying until more than 3 months of age (Olitsky et al 2011).

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